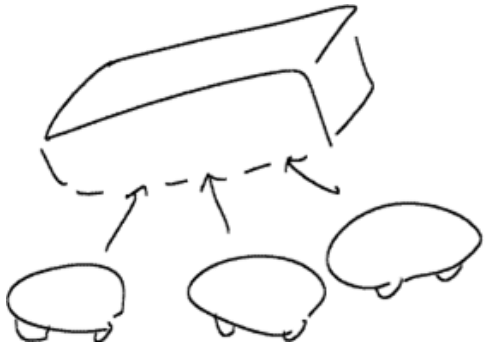
Multi-touch Interface for Controlling Multiple Mobile Robots

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JST, ERATO, IGARASHI Design UI Project
Jun Kato

INTRODUCTION

Motivation

- Multiple mobile robots can do various tasks with greater efficiency.





- They also improve fault tolerance.



- Then, how would you like to control movements of those robots?
  - “Discussion of Challenges for User Interfaces in Human-Robot Teams” - (Driewer, F., 2007)
  - “Human control for cooperating robot teams” – (Wang, J., 2007)

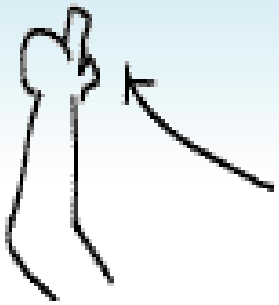
Existing User Interfaces



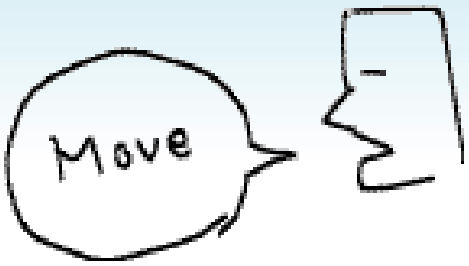
Joystick



Mouse



Gesture



Speech

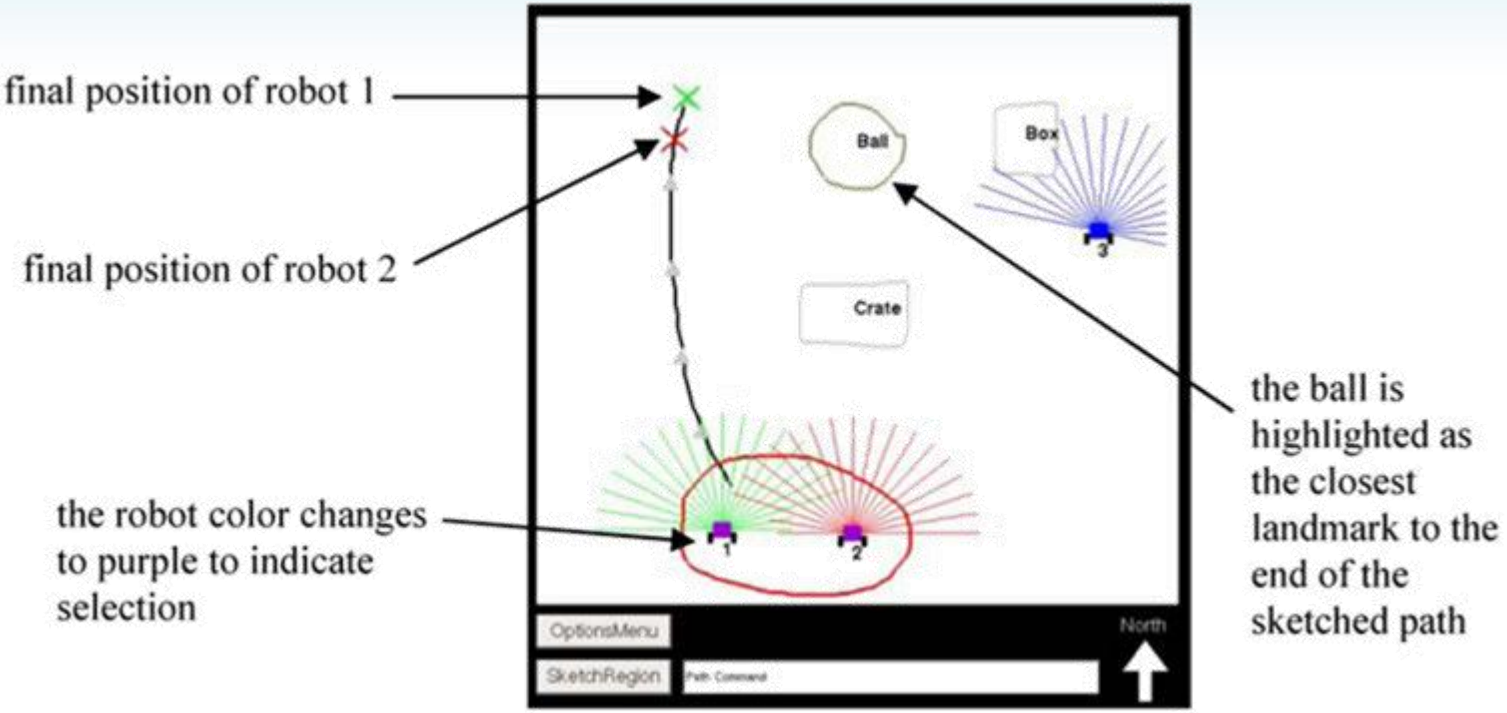


Gesture and Speech (Rogalla, 2002)



PDA and Pen (Fong, T., 2002)

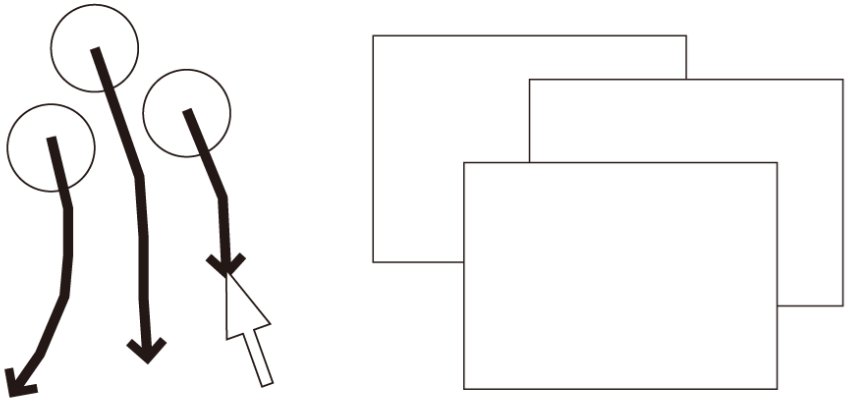
Existing User Interfaces



Drawing a sketch to control robots (Skubic, M., 2007)

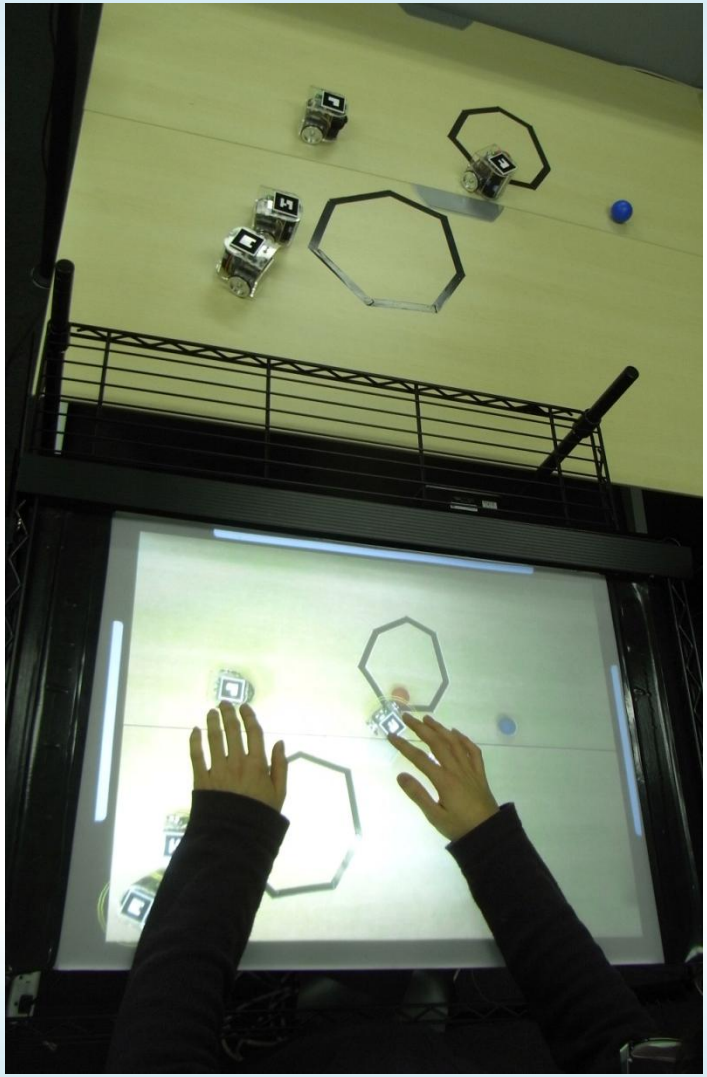
Problems

- Draw similar paths? Switch among many views?

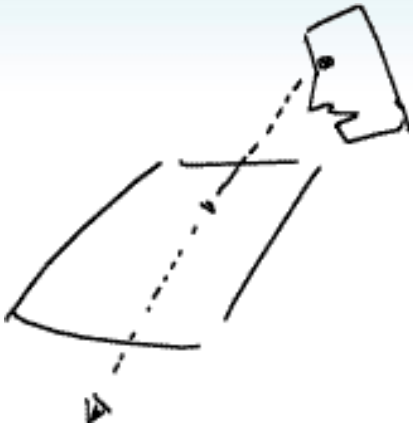


- How can we combine these interfaces with autonomous approaches?

My Approach



God's view of the environment



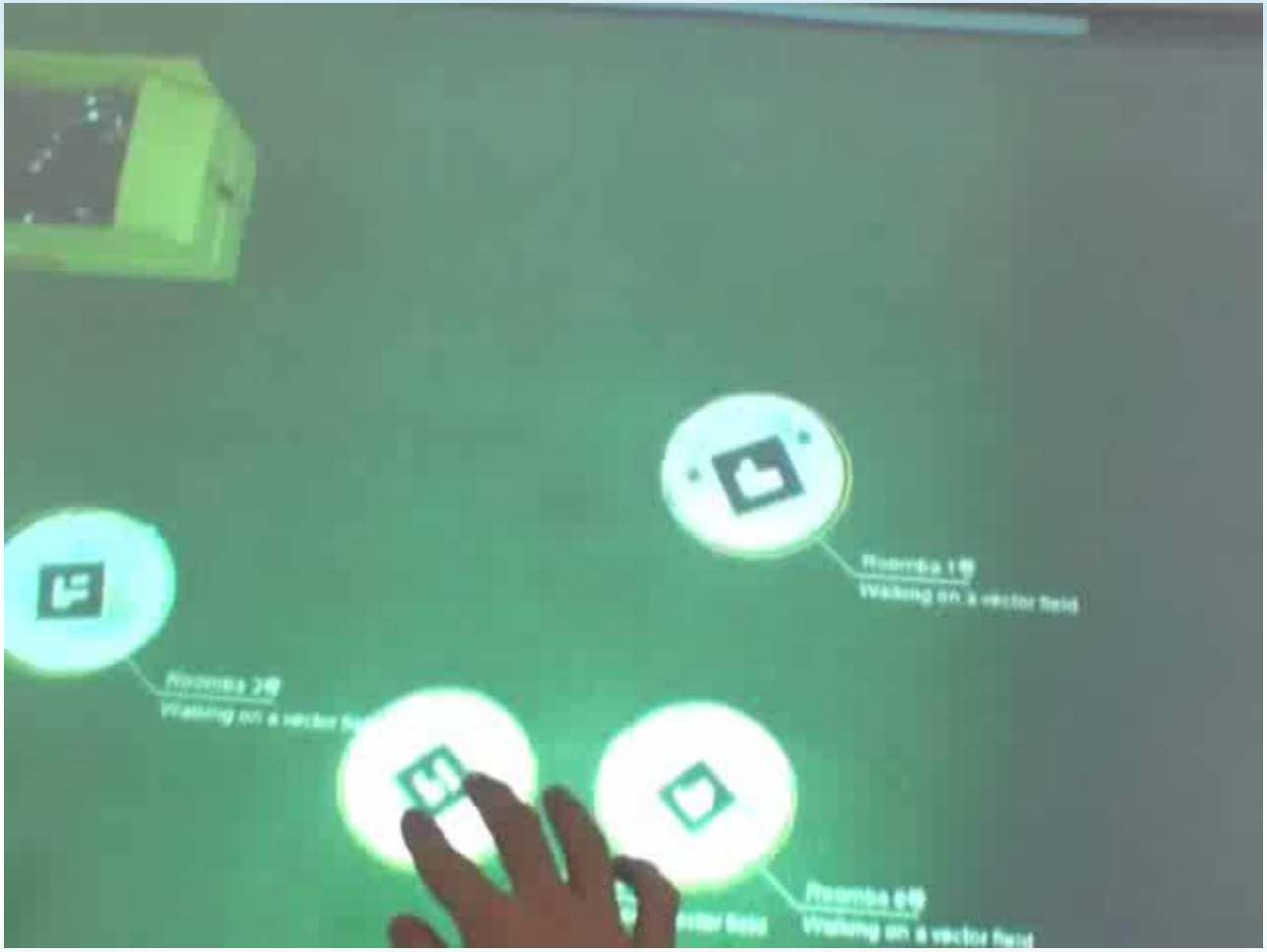


Two hands' intuitive operation

↓

Direct manipulation of raw data
for navigating robots

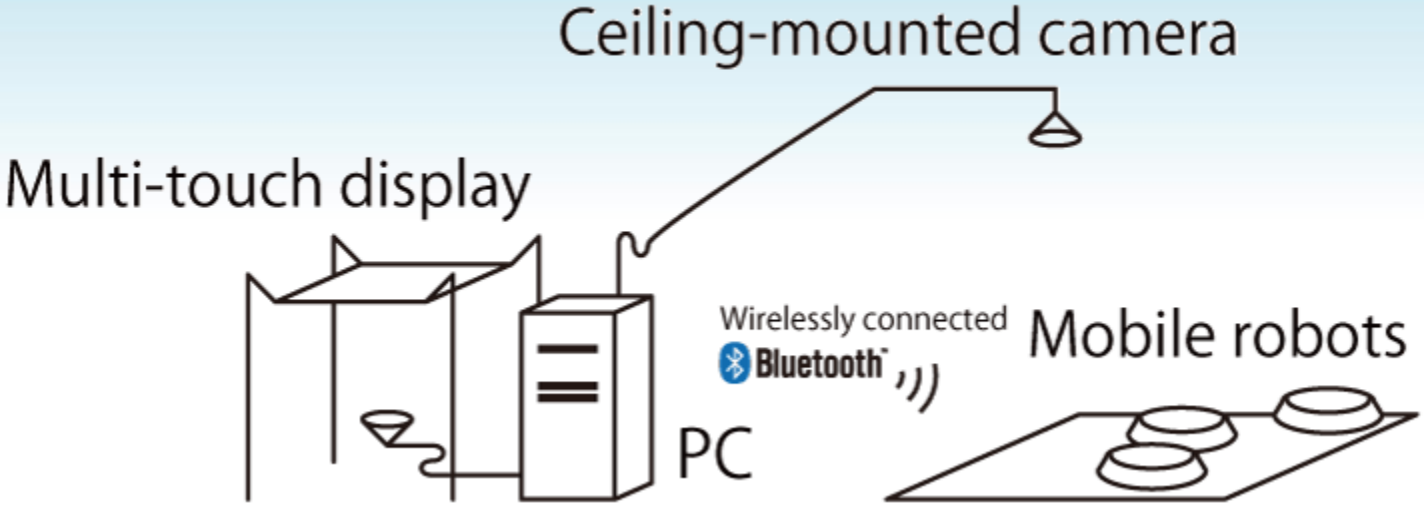
Vector Field Operation



Draw a stream, drift robots!

VECTOR FIELD OPERATION

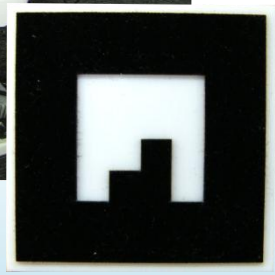
Hardware Setup



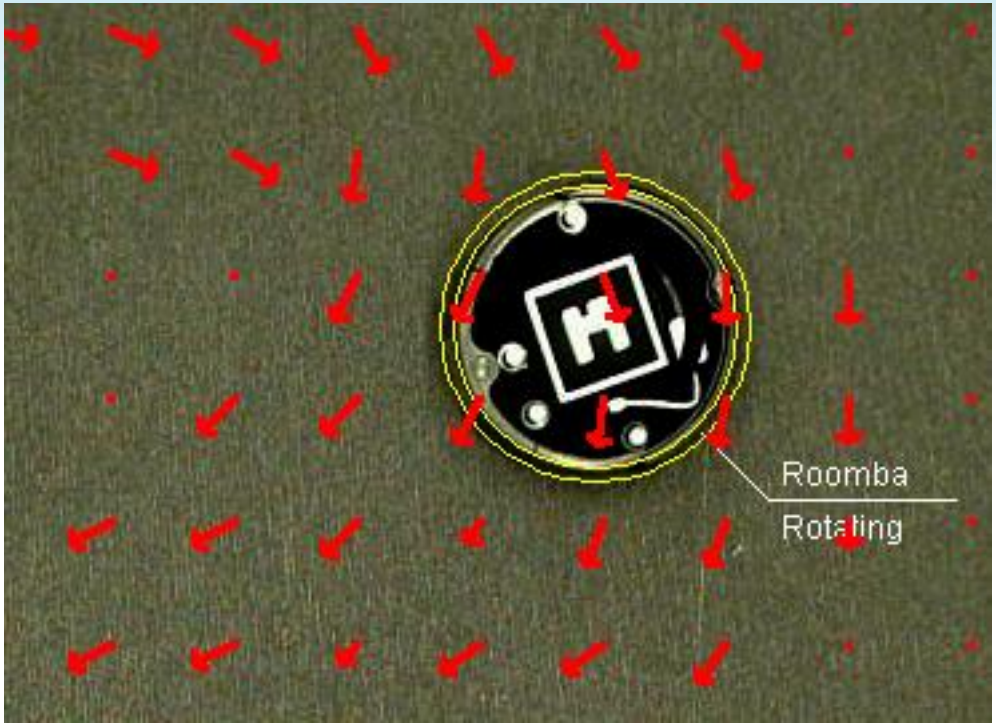




with



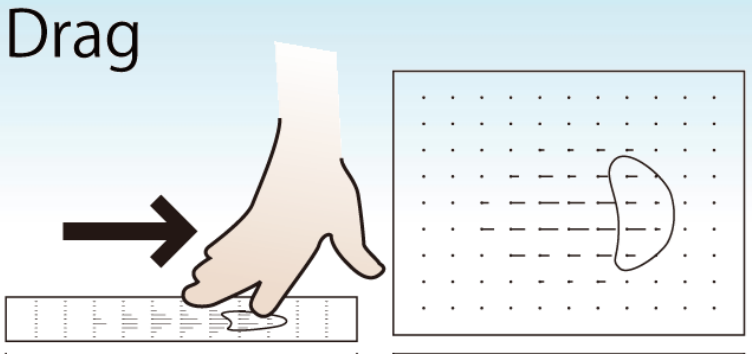
Vector Field on the View



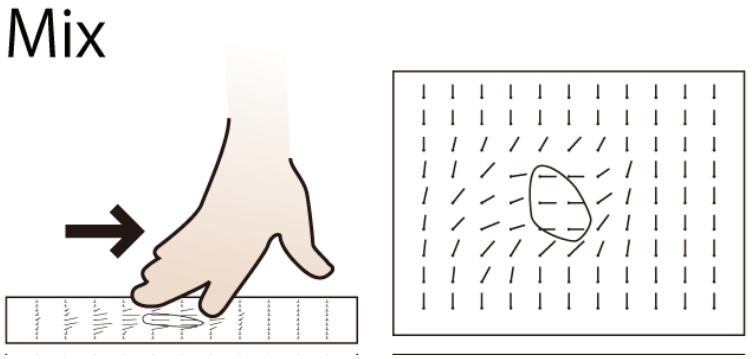
- The view is divided into grids.
- Each grids have 2D vector information.
- Whole grids construct a 2D vector (flow) field.

Available Operations on the Vector Field

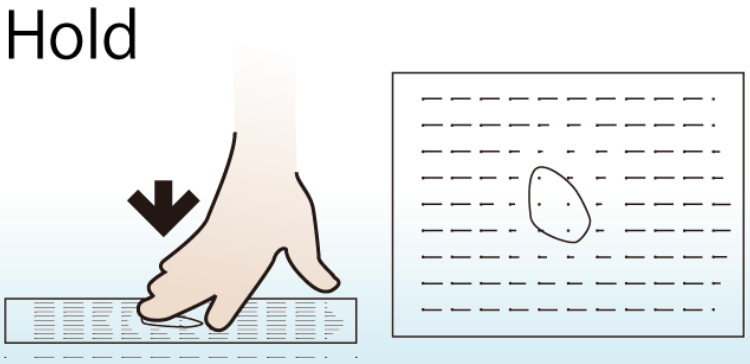
To move robots,



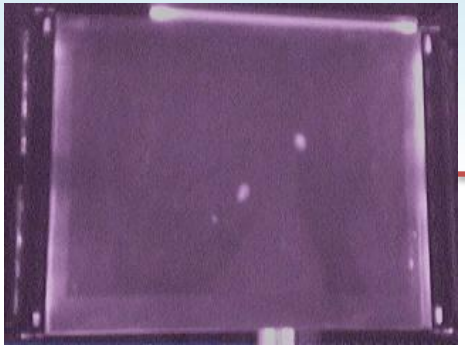
To fix movements,



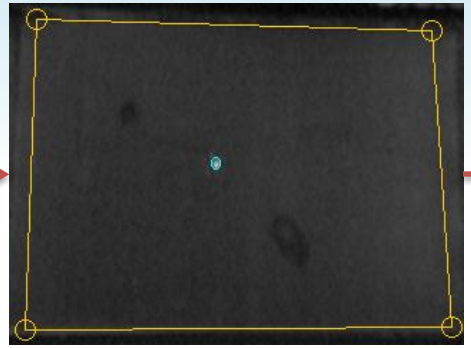
To stop robots,



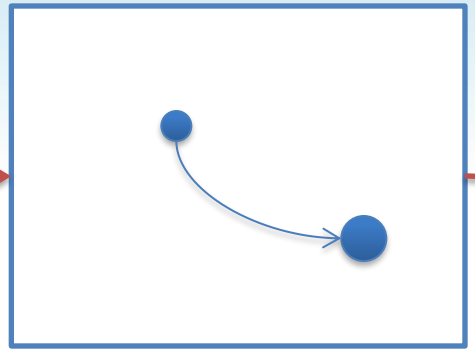
Implementation of the Vector Field



Capture

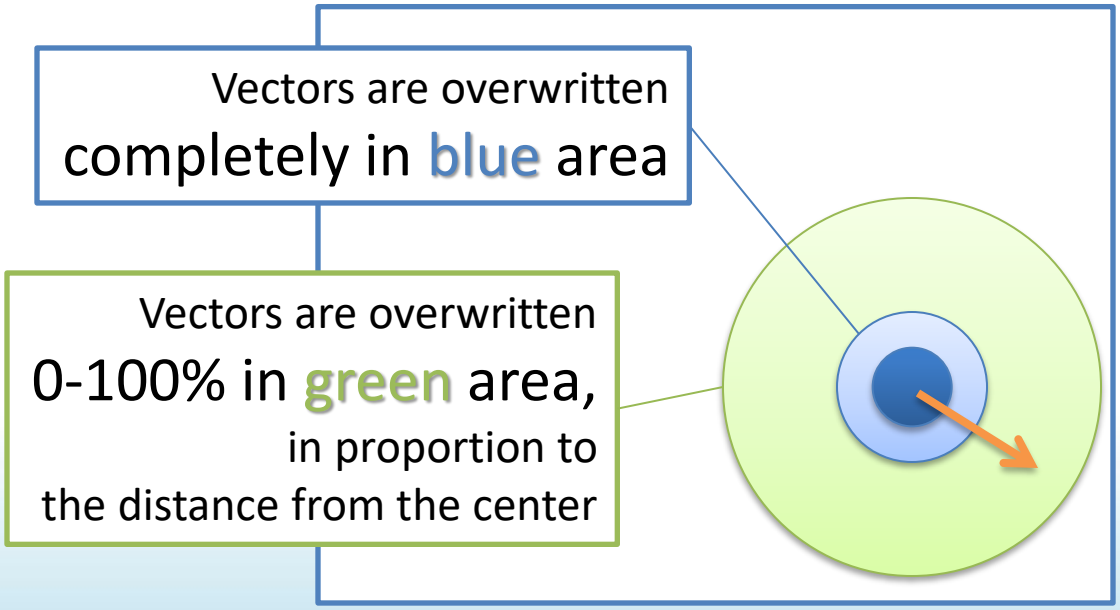


Calibrate

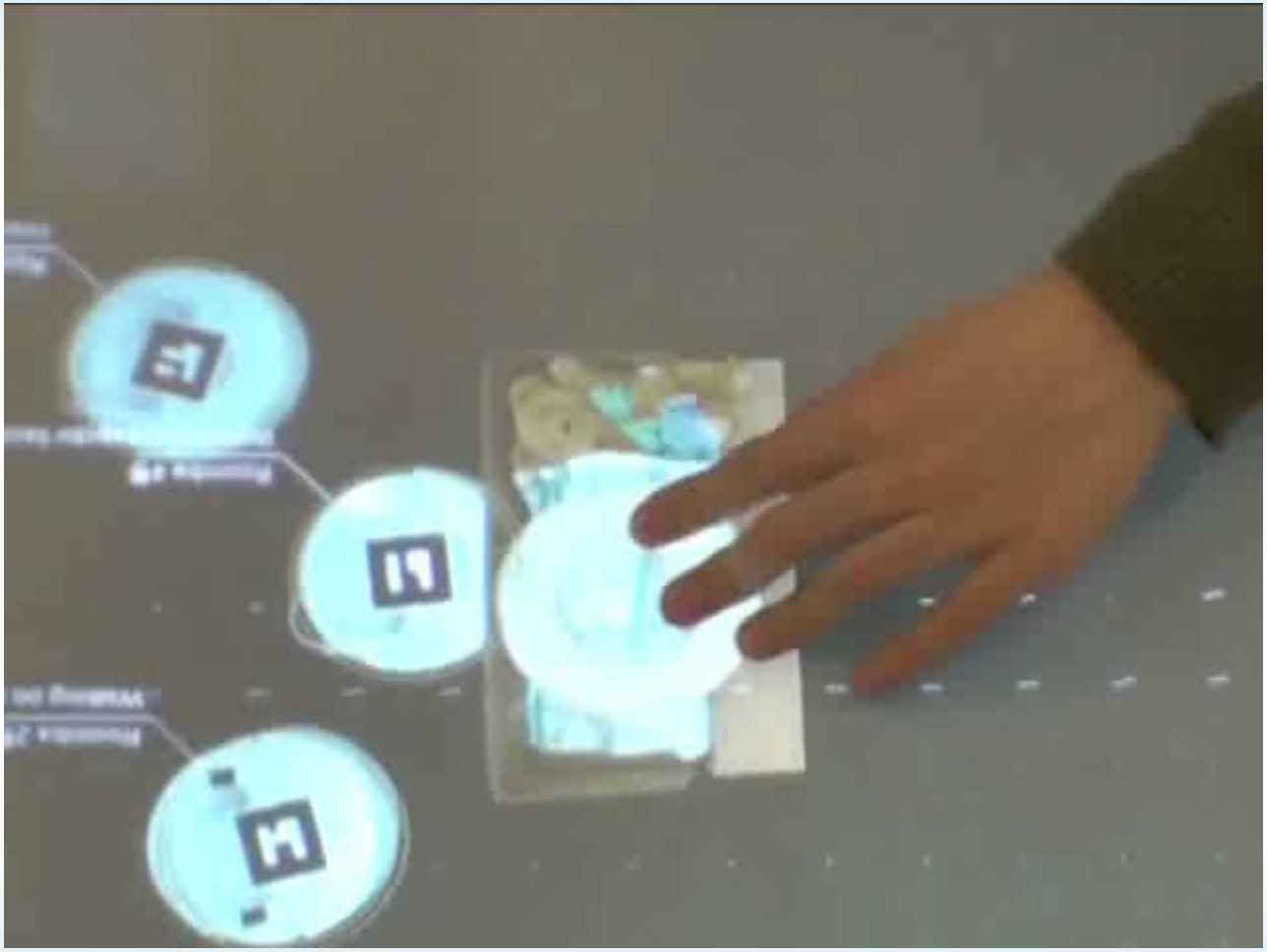


Track motion

Motion vector affects the field



“So, what can we do?”

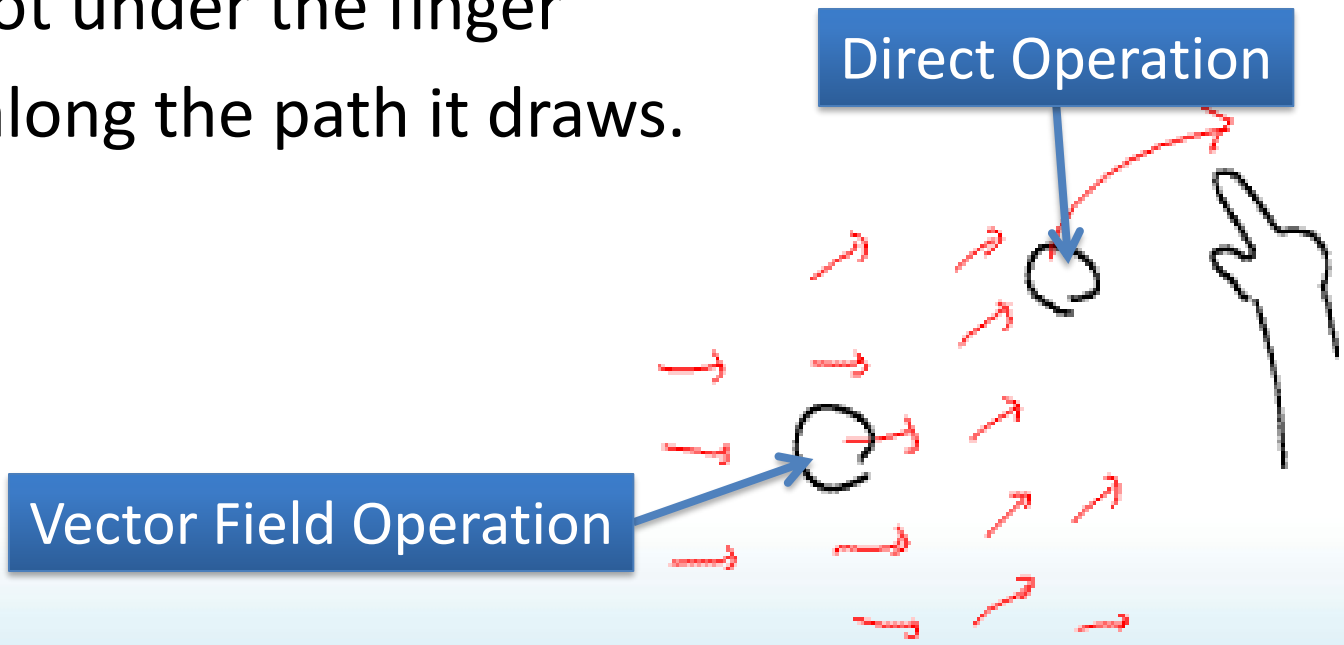


Next Step…

- A user test
- Integration of other user interfaces

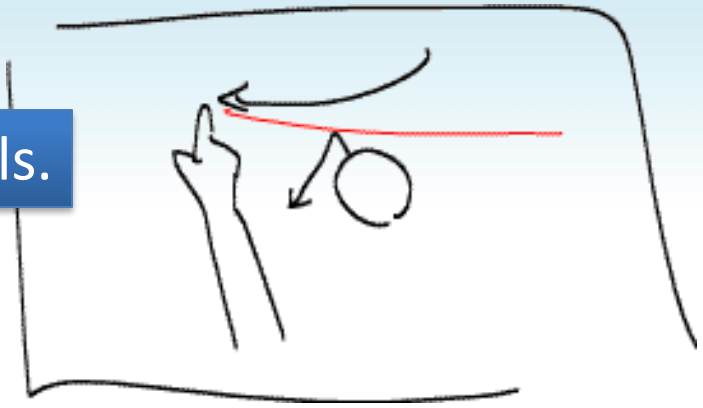
Combination with Direct Operation

- Based on Vector Field Operation
- When fingers are in the robot icon, Direct Operation starts.
  - The robot under the finger moves along the path it draws.

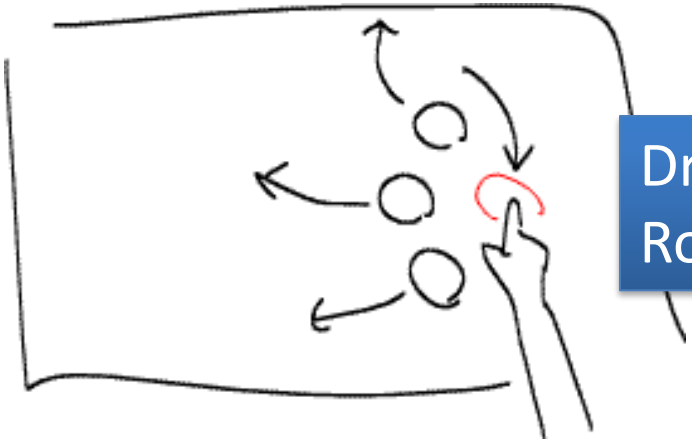


Extensions of Vector Field Operation

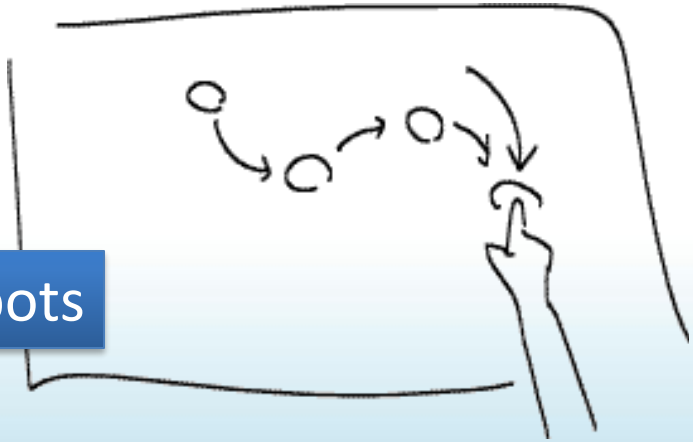
Draw or clear virtual walls.



Draggable virtual dog icons.
Robots as sheeps run away!



Bind relative positions of robots

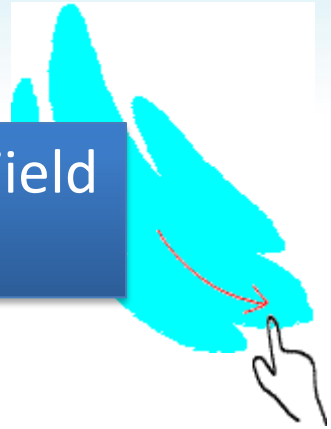


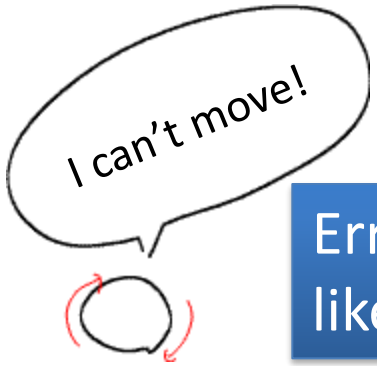
Integration with Autonomous Algorithms

- With Virtual Force Field (Borenstein, J., 1989)
- Etc.?

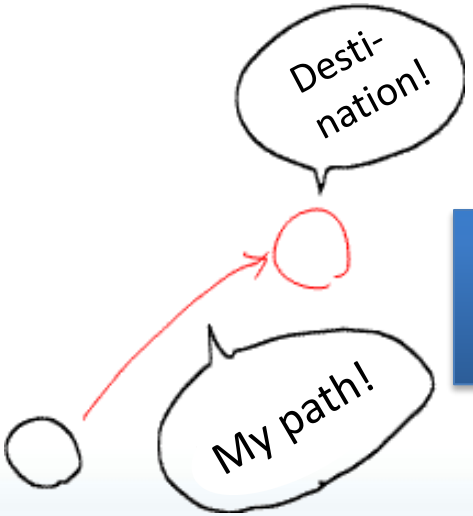
Extensions of Visualization

Visualization of the Vector Field with particle-animation





Error displays like time out



Path calculation and visualization

Summary

- We developed a multi-touch interface for controlling multiple mobile robots simultaneously.
- Our interface has capability to be integrated with other operating methods, including autonomous ways.

